

Shounak Nandi

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EDUCATION

Master of Science in Biomedical Engineering and Medical Physics

Apr. '21 – Apr. '24

Technical University of Munich, GPA: 3.23/4.00

Munich, Germany

Bachelor of Technology in Biomedical Engineering

Aug. '13 – Jun. '17

Maulana Abul Kalam Azad University of Technology, GPA: 3.98/4.00

Kalyani, India

RESEARCH EXPERIENCE

Study Coordinator

Jun.'24 – present

Radiology (PI: Tim Duong), Albert Einstein College of Medicine

New York City, USA

- Perform pre- and post-processing of multi-modal magnetic resonance imaging (MRI) data, including diffusion MRI, functional MRI, arterial spin labeling, structural T1w and FLAIR images
- Utilize neuroimaging tools (FSL, MRtrix, Freesurfer, SPM, DIPY, AFNI) to create specialized processing pipelines for data analysis
- Build pipelines for processing big-data with use of a High-Performance Compute cluster for data analysis
- Analysis, organization, and research of clinical data and cognitive batteries for multiple studies
- Maintain regulatory compliance and documentation, assist in designing study protocols for NIH research
- Collaborate with faculty researchers and mentor high-school students for research projects in the field of oncology and neurodegenerative diseases

Visiting Scholar

Jan. – May. '24

Accessible MR Lab (PI: Sairam Geethanath), Johns Hopkins University School of Medicine

Baltimore, USA

- Analyze and optimize the low-resolution MR images from very low-field scanner, volumetric and textural analysis on substance-use-disorder patients, building classifier models to predict treatment responders

Associate Research I

Apr. – Dec. '23

Accessible MR Lab (PI: Sairam Geethanath), Icahn School of Medicine at Mount Sinai

New York City, USA

- Analyze and optimize the low-resolution MR images from very low-field scanner, volumetric and textural analysis on substance-use-disorder patients, building classifier models to predict treatment responders
- Use segmentation algorithms (FSL, FreeSurfer, ANTS) to perform segmentation of the brain volumes
- Perform data collection and quality control of study data, localization of the brain subcortical regions during the MR scan, and further compute MR spectroscopy and data analysis for opioid-use-disorder patients (Dept. of Psychiatry, ISMMS PI: Yasmin Hurd)

Student Research Assistant

Apr. – Oct. '22

Interdisciplinary Research Lab (PI: Nassir Navab), Technical University of Munich

Munich, Germany

- DigiBiop- Built a Python project for automatically sorting multi-modal (PET-MR) prostate DICOM images based on the type of scan and other patient demographics
- Real-time registration of pre-operative computed tomography (CT) data with inter-operative Ultrasound data- Worked on building an algorithm that calibrates the point cloud of the real-world phantom with third-party imaging software to perform real-time optical tracking, using C++ and Python
- Acquired and analyzed 6D optical tracking data measurements and calibrated the optical tracker with the wire and body phantoms using optical markers on the same software

JOURNAL PUBLICATIONS

Published

- **S. Nandi**, P. Poojar, S. Taufique, et al., “Magnetic Resonance Imaging Textural Changes Are More Sensitive Than Volumetric Changes in the Amygdala of Cocaine Use Disorder Patients,” *NMR in Biomedicine* 39, no. 1 (2026): e70193, <https://doi.org/10.1002/nbm.70193>.
- McKenna F*, **Nandi S***, Henry SS, Nayak S, Fleysher R, Duong TQ. Microvascular and microstructural brain abnormalities in paediatric haematological cancer survivors are related to cognitive deficits: An IVIM-FWI MRI study. *Br J Haematol.* 2026; 00: 1–12. <https://doi.org/10.1111/bjh.70300> *co-first authors.

- Nguyen HQ, Hadidchi R, Eligulashvili A, **Nandi S**, Vichare A, Patel B, Sanchez J, Bisulca J, Henry S, Lee JS, Duong TQ. COVID-19 and radiological progression of multiple sclerosis. *Diagnostics*, 16(10), 1513. <https://doi.org/10.3390/diagnostics16101513>

Submitted/Under Review

- Henry SS, **Nandi S**, Holtzer R, Rabin M, Gennarini L, Tal A, Lu J, Duong TQ. Chemotherapy-induced changes in temporal lobe cortical thickness of pediatric cancer survivors in a diverse urban population; *Br J Haematol*. 2026
- Nayak S, **Nandi S**, McKenna F, Henry SS, Duong TQ. Cognitive Control in Pediatric Cancer Survivors: A task-fMRI study; *Br J Haematol*. 2026
- Muir ER*, **Nandi S***, Vichare A, Henry SS, Duong TQ. Neuroimaging Correlates of Chemotherapy-Related Cognitive Impairment in Pediatric Hematologic Cancer Survivors: A Systematic Review; *Pediatr Blood Cancer*. 2026. *co-first authors

CONFERENCE ABSTRACTS & PRESENTATIONS

International

- **Nandi S**, Poojar P, Bachi K, Taufique S, Hurd YL, Geethanath S. Textural changes in the amygdala are more sensitive than volumetric changes in cocaine use disorder patients undergoing therapy. *Proc. Intl. Soc. Mag. Reson. Med.*, Singapore. 2024. Poster Presentation.
- Srinivas SA, Alon L, Alipour A, **Nandi S**, Purchase A, et al. MRI4ALL: A week-long hackathon for the development of an open-source ultra-low-field MRI system. *Proc. Intl. Soc. Mag. Reson. Med.*, Singapore. 2024. Oral Presentation.
- **Nandi S**, Poojar P, Bachi K, Taufique S, Hurd YL, Geethanath S. Textural changes are more sensitive than volumetry of amygdala in cocaine use disorder patients: potential for low field translation. *2nd ISMRM Workshop on Accessible MRI*, New Delhi, India. 2024. Poster Presentation.
- **Nandi S**, McKenna FF, Nayak S, Henry S, Fleysher R, Duong TQ. Cerebellar–cortical white-matter disorganization is associated with cognitive impairment in pediatric cancer survivors: A NODDI-Bingham study. *SOBP Annual Meeting*, New York, USA. 2026. Poster Presentation.
- **Nandi S**, Geethanath S. LASSO-informed textural changes are more sensitive than volumetry of amygdala in cocaine use disorder patients. *Intl. Soc. Mag. Reson. Med.*, Cape Town, South Africa. 2026. Poster Presentation.
- Nayak S, **Nandi S**, McKenna FF, Duong TQ. Task-based deactivations in a fast event-related fMRI design are better captured by deconvolution than FIR basis functions. *Intl. Soc. Mag. Reson. Med.*, Cape Town, South Africa. 2026. Power-Pitch Presentation.
- McKenna FF, **Nandi S**, Nayak S, Henry S, Fleysher R, Duong TQ. Altered IVIM-FWI metrics in pediatric hematologic cancer survivors and their relationship with cognitive deficits. *Intl. Soc. Mag. Reson. Med.*, Cape Town, South Africa. 2026. Poster Presentation.

Domestic

- Poojar P, **Nandi S**, Bachi K, Gardner G, Oort JV, Zadeh EM, Geethanath S, Hurd YL. Spectral quality and localization in MR spectroscopy data from opioid use disorder patients. *Annual Neuroscience Retreat*. New York, USA. 2025. Poster Presentation.

INDUSTRY EXPERIENCE

Regional Service Engineer (Field-service)

Jul. '18 – Apr. '21

Karl Storz Endoskope

Kolkata, India

- Supported the surgeons and support staff by providing on-field solutions (also during surgeries), preliminary troubleshooting of the devices, level 2 breakdown repairs, user application training, installation, and preventive maintenance
- Additional role- Interaction with corporate stakeholders and KOLs for negotiation and product promotion, team handling, managing installation base (service support) for more than 150 customers
- Units handled- Laparoscopes, Modular Operating rooms, Laser system, Surgical hand instruments

Trainee Sales and Service Engineer

Jun. '17 – Jul. '18

Alliance Biomedica

Kolkata, India

- Supported the physicians and healthcare team with on-field solutions, including preliminary device troubleshooting, level 1 breakdown repairs, user application training, installation, preventive maintenance and consumables sales
- Units handled- EEG, EMG, Surgical aspirators, Audiometer, Tympanometer, Nebulizers

TRAINING AND INTERNSHIPS

MRI4ALL Hackathon | *MRI, Research, Troubleshooting, Programming*

New York University

Oct. '23

New York City, USA

- Participated in a 5-day hackathon to build a completely open-source very low-field MR scanner, Zeugmatron Z1 from scratch. The scanner has ~ 1000 permanent magnets, active and passive shimming with in-built liquid-cooling technology. <https://doi.org/10.1002/jmri.29771>

Innovation Think Tank Certification Program | *Innovation, time management, teamwork*

Siemens Healthineers and Fakeeh University Hospital

Jan. '22

Dubai, UAE

- Participated in a 9-day co-creation and innovation program with a global team to propose a solution after considering the KOL voices on the trending challenges to shape the future of healthcare of the UAE

Research Internship | *Research, Literature survey, EEG, LabView, Analysis, design*

National Institute of Technology

Jun. – Jul. '16

Raipur, India

- Researched the effect on Human EEG due to word stimulus and designed an experiment based on event-related potential (ERP) for EEG Data acquisition (real-time)
- Project- “Emotion and attention-based EEG experiment design and analysis”

Hospital Internship (Biomedical dept.) | *Maintenance, Troubleshooting, Calibration*

Mission of Mercy Hospital and Research Centre

Apr. – May '16

Kolkata, India

- Tasks: Overall breakdown maintenance and service of medical equipment, calibration, and preventive maintenance
- Assisted in the equipment purchase plan and maintaining vendor correspondence of the hospital's medical devices

Research Internship | *Research, Literature survey, Market survey, Waste treatment design*

Indian Institute of Technology

Jun. – Jul. '14

Kharagpur, India

- Researched on the waste recycling industry. Marketed the concept of vermicompost
- Designed a waste treatment model for the IIT Kharagpur campus
- Project- “Innovation Challenge Grant Under Green Policy Initiative for IIT Kharagpur campus”

TECHNICAL SKILLS

Programming Skills: Python, C++, MATLAB, LaTeX, Shell, R, SQL

Neuroimaging tools: FSL, FreeSurfer, AFNI, jMRUI, ANTs, SPM

Developer Tools: Git, VS Code, PyCharm, Anaconda, MS Office, HPC, Slurm

Libraries and framework: Pytorch, NumPy, Matplotlib, Scipy, Pydicom, Dipy, Csv, Pandas, Mahotas, Seaborn

Platform: Windows, Linux, Darwin

Engineering: MRI, MR Spectroscopy, Computed tomography, X-ray, Image processing, Multimodal imaging, medical devices, Machine Learning and Deep Learning, Prompt Engineering, AI tools

LANGUAGE SKILLS

Bengali (native), **English** (fluent), **Hindi** (native), **German** (intermediate)

SOFT SKILLS

Public Speaking & Training, Client Relations, Troubleshooting, Decision Making, Independent & Self-Driven, Adaptability

AWARDS & RECOGNITIONS

- **ISMRM Trainee Stipend** - Awarded for getting selected to present scientific work at the Annual conference of ISMRM 2024.
- **Master Thesis Travel Grant** - Awarded for getting selected for the International conference at the ISMRM'2024 during Master thesis project
- **JIS College Blue 2k17** (Best student award)- Awarded to the best student for all-round performance.
- **JIS Tech Symposium 2k17**- Best Project award for "Patient safety using an isolation transformer".
- **JIS Samman** (Departmental topper 2014, 2015, 2016)- Awarded for securing the first rank in the department.

CONFERENCE AND CO-CURRICULAR ACTIVITIES

- **i2i Workshop, New York**: Poster and Power pitch- Presented my work on volumetric and textural analysis on cocaine use disorder patients (2023)
- **Community Manager of DEGIS- International Students community, Munich city**. My role was to connect with the community and address their problems while living in Munich as a student (Jan-May 2023)
- **Guest vlogger** in an episode on "a Day in Life of a Student in Germany" for a YouTube channel named Bharat in Germany (350K+ Subscribers) (January 2022)
- Invited as a **guest speaker** for "Introduction to the world of Endoscopy" at the webinar series: Session I of Core Biomedical Engineering Concepts organized by Bee Biomed (2020)
- Exhibited and volunteered in the International Book Fair 2016, and other technical symposiums and conferences
- Worked as an Asst. Joint Secretary- "Creatively Yours" (Language and Literary Club of JIS College of Engineering)
- Campus representative of Force Biomedical, an all-India biomedical engineers' community, for JIS College of Engineering

PERSONAL INTERESTS

Music, photography and videography, cooking and baking, Table tennis, Badminton.

References are available upon request.